

Abstract

This project is a lightweight web application to digitize hostel records. The Admin can add students (with their hostel and room), manage hostels, and publish notices either for a specific hostel or all hostels. Students log in with their Student ID to view their details, read notices, and submit Maintenance and Leave applications. For Maintenance/Leave, the system auto-fills the hostel from the student's stored data—students cannot change it. The stack is HTML/CSS/JS (frontend), Node.js + Express (backend), and MySQL (database).

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Introduction

Manual hostel record-keeping leads to errors, duplication, and slow retrieval. The goal is to build a simple web app that:

- Centralizes student and hostel data.
- Allows students to self-serve (login, view details, submit requests).
- Lets Admins post notices to one hostel or all.
- Ensures Maintenance and Leave requests are tied to the **correct hostel automatically**.

This version removes the rooms/allotment module: **Admin directly saves student details** (Student ID, Name, Phone, Hostel Name, Room Number).

Project Objectives

- Provide **Admin Portal** to add students and hostels, and post notices (hostel-specific or global).
- Provide **Student Portal** to login via Student ID, view profile and notices, and submit Maintenance/Leave forms.
- Auto-bind Maintenance/Leave requests to the student's hostel (no manual hostel selection).
- Keep the design small, understandable, and deployable on free tiers.

System Requirements

Hardware Requirements

Component Minimum Requirement Recommended

Processor	Dual Core 2.0 GHz	Intel i5 / Ryzen 5
RAM	4 GB	8 GB
Storage	500 MB	2 GB free
Display	1024 × 768	1920 × 1080

Software Requirements

Software	Version / Details
Operating System	Windows / Linux / macOS
Database	MySQL 8.0+
Database Tool	MySQL Workbench (optional)
Backend	Node.js + Express.js
Frontend	HTML, CSS, JavaScript
Browser	Chrome / Edge / Firefox
Editor	VS Code / Sublime / Notepad++

Functional Requirements

- Admin Login & Dashboard
- Add and View Hostels
- Add and View Students (Student ID, Name, Phone, Hostel, Room)
- Post Notices for:
 - All hostels
 - Specific hostel
- Student Login using **Student ID**
- Students can submit:
 - Maintenance Requests (hostel auto-detected)
 - Leave Applications (hostel auto-detected)
- Students can view status of their requests
- Students can see notices relevant to their hostel

Non-Functional Requirements

- Data Consistency (Foreign Keys)
- Secure Access (Role-based: Admin / Student)
- Scalable & Maintainable Schema
- Easy to use User Interface

System Design

ER Diagram

[HOSTELS] 1 ———— < ∞ [STUDENTS]

id (PK)

name (UNIQUE)

address

student_id (PK)

name

phone

room_number

hostel_id (FK → HOSTELS.id)

[HOSTELS] 1 ———— < ∞ [NOTICES]

id

id (PK)

title

body

audience ('all'/'hostel')

hostel_id (nullable FK)

created_at

[STUDENTS] 1 ———— < ∞ [MAINTENANCE_REQUESTS]

student_id

id (PK)

student_id (FK → STUDENTS.student_id)

hostel_id (FK → HOSTELS.id) auto

attached

category

description

status

created_at

[STUDENTS] 1 —————< ∞ [LEAVE_APPLICATIONS]

student_id

id (PK)

student_id (FK → STUDENTS.student_id)

hostel_id (FK → HOSTELS.id) auto-
attached

from_date

to_date

reason

status

applied_at

[ADMINS]

id (PK)

username (UNIQUE)

password

Details of Tables (Schema Description)

Table Name	Purpose
admins	Stores admin login credentials
hostels	Stores hostel names & addresses
students	Stores student personal & hostel room details
notices	Stores announcements (global or hostel-specific)
maintenance_requests	Stores maintenance complaints submitted by students
leave_applications	Stores leave requests submitted by students

Implementation

Database Creation (SQL)

```
DROP DATABASE IF EXISTS hostel_database;
```

```
CREATE DATABASE hostel_database;
```

```
USE hostel_database;
```

```
CREATE TABLE admins (
```

```
    id INT AUTO_INCREMENT PRIMARY KEY,
```

```
    username VARCHAR(100) UNIQUE,
```

```
    password VARCHAR(255)
```

```
);
```

```
INSERT INTO admins (username, password) VALUES ('radhey', 'god');
```

```
CREATE TABLE hostels (
```

```
    id INT AUTO_INCREMENT PRIMARY KEY,
```

```
    name VARCHAR(150) UNIQUE,
```

```
    address VARCHAR(255)
```

```
);
```

```
CREATE TABLE students (
```

```
    student_id VARCHAR(20) PRIMARY KEY,
```

```
    name VARCHAR(200),
```

```
    phone VARCHAR(20),
```

```
    room_number VARCHAR(20),
```

```
    hostel_id INT,
```

```
    FOREIGN KEY (hostel_id) REFERENCES hostels(id)
```

```
);
```

```

CREATE TABLE notices (
    id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(200),
    body TEXT,
    audience ENUM('all','hostel'),
    hostel_id INT NULL,
    FOREIGN KEY (hostel_id) REFERENCES hostels(id)
);

CREATE TABLE maintenance_requests (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id VARCHAR(20),
    hostel_id INT,
    category VARCHAR(100),
    description TEXT,
    status ENUM('open','in_progress','closed') DEFAULT 'open',
    FOREIGN KEY (student_id) REFERENCES students(student_id),
    FOREIGN KEY (hostel_id) REFERENCES hostels(id)
);

CREATE TABLE leave_applications (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id VARCHAR(20),
    hostel_id INT,
    from_date DATE,
    to_date DATE,
    reason VARCHAR(255),

```

```
status ENUM('pending','approved','rejected') DEFAULT 'pending',  
FOREIGN KEY (student_id) REFERENCES students(student_id),  
FOREIGN KEY (hostel_id) REFERENCES hostels(id)  
);
```

Frontend (Screens Overview)

Page Name	Purpose
admin-login.html	Admin authentication
admin-dashboard.html	Add hostels & students
notices.html	Post and view notices
student-login.html	Student authentication
student-dashboard.html	View profile & notices
maintenance.html	Submit maintenance requests
leave.html	Submit leave application

Backend (Node.js Express APIs) – Explanation

Endpoint	Function
POST /api/admin/login	Admin authentication
POST /api/hostels / GET /api/hostels	Manage hostels
POST /api/students / GET /api/students	Manage students
POST /api/notices / GET /api/notices	Post and display notices
POST /api/maintenance	Submit maintenance (hostel auto-filled)
GET /api/maintenance/mine/:id	Student's maintenance history
POST /api/leaves	Submit leave (hostel auto-filled)
GET /api/leaves/mine/:id	Student's leave history

```
mysql> SELECT * FROM STUDENTS;
```

student_id	name	phone	hostel_id	room_number
S1001	Amit Kumar	9000000001	1	101
S1002	Priya Sharma	9000000002	1	102
S1003	Raj Verma	9000000003	2	201
S1004	Radhey	9306907930	1	D-318

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM ADMINS;
```

id	username	password
1	radhey	god

```
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM HOSTELS;
```

id	name	address
1	Shivaji Boys Hostel	CSJMU
2	Sawran Boys Hostel	CSJMU

```
2 rows in set (0.00 sec)
```

```
mysql> DESC STUDENTS;
```

Field	Type	Null	Key	Default	Extra
student_id	varchar(20)	NO	PRI	NULL	
name	varchar(200)	NO		NULL	
phone	varchar(20)	YES		NULL	
hostel_id	int	NO	MUL	NULL	
room_number	varchar(20)	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql> DESC maintenance_requests;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
student_id	varchar(20)	NO	MUL	NULL	
hostel_id	int	NO	MUL	NULL	
category	varchar(100)	NO		NULL	
description	text	NO		NULL	
status	enum('open','in_progress','closed')	NO		open	
created_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

```
7 rows in set (0.00 sec)
```

```
mysql> DESC NOTICES;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
body	text	NO		NULL	
audience	enum('all','hostel')	NO		all	
hostel_id	int	YES	MUL	NULL	
created_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

```
6 rows in set (0.00 sec)
```

```
mysql> DESC LEAVE_APPLICATIONS;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
student_id	varchar(20)	NO	MUL	NULL	
hostel_id	int	NO	MUL	NULL	
from_date	date	NO		NULL	
to_date	date	NO		NULL	
reason	text	NO		NULL	
status	enum('pending','approved','rejected')	NO		pending	
applied_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

```
8 rows in set (0.01 sec)
```

```
mysql> DESC ADMINS;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
username	varchar(100)	NO	UNI	NULL	
password	varchar(255)	NO		NULL	

```
3 rows in set (0.02 sec)
```

```
mysql> DESC HOSTELS;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
name	varchar(200)	NO	UNI	NULL	
address	varchar(255)	YES		NULL	

```
3 rows in set (0.00 sec)
```



```
mysql> USE hostel_database;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_hostel_database |
+-----+
| admins                     |
| hostels                   |
| leave_applications        |
| maintenance_requests      |
| notices                   |
| students                  |
+-----+
6 rows in set (0.02 sec)
```

```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| hostel_database |
| hostel_mgmt    |
| information_schema |
| mysql          |
| performance_schema |
| sakila         |
| sys            |
| world          |
+-----+
8 rows in set (0.02 sec)
```

Admin Portal

Admin Dashboard - Students

Notices

Maintenance

Leaves

Refresh

Logout

Add Student

Student ID

Name

Phone

Select Hostel

Room Number

Add

All Students

ID	Name	Phone	Hostel	Room	Actions	
S1003	Raj Verma	9000000003	Sawran Boys Hostel	201	Delete	Edit
S1001	Amit Kumar	9000000001	Shivaji Boys Hostel	101	Delete	Edit
S1002	Priya Sharma	9000000002	Shivaji Boys Hostel	102	Delete	Edit
S1004	Radhey	9306907930	Shivaji Boys Hostel	D-318	Delete	Edit

Post Notice

Title

Body

All Hostels

Select Hostel

Publish

Leave Applications (Admin)

Filter by Student ID

All statuses

Refresh

1 shown / 1 total

ID	Student ID	Student Name	Hostel	From	To	Reason	Status	Applied At	Actions
1	S1004	Radhey	Shivaji Boys Hostel	2025-11-08T18:30:00.000Z	2025-11-16T18:30:00.000Z	need to go home	pending	2025-11-09T16:52:10.000Z	Approve Reject Reset

Post Notice

Title

Body

All Hostels

Select Hostel

Publish

Maintenance Tickets

ID	Student	Room	Category	Description	Status	Actions
1	Radhey	-	No Electricity	please maintain them	in_progress	OpenIn ProgressClose

Maintenance Tickets

ID	Student	Room	Category	Description	Status	Actions
1	Radhey	-	No Electricity	please maintain them	in_progress	OpenIn ProgressClose

Leave Applications (Admin)

Filter by Student ID All statuses Refresh 1 shown / 1 total

ID	Student ID	Student Name	Hostel	From	To	Reason	Status	Applied At	Actions
1	S1004	Radhey	Shivaji Boys Hostel	2025-11-08T18:30:00.000Z	2025-11-16T18:30:00.000Z	need to go home	pending	2025-11-09T16:52:10.000Z	Approve Reject Reset

Student Portal

My Details

ID: S1004

Name: Radhey

Phone: 9306907930

Hostel: Shivaji Boys Hostel

Room: D-318

Notices

Maintenance

Leave

Logout

Maintenance Request

Submit

My Tickets

ID	Category	Description	Status	Created
1	No Electricity	please maintain them	in_progress	09/11/2025, 22:22:44

Leave Application

dd-mm-yyyy

dd-mm-yyyy

Reason

Apply

My Leave Applications

ID	From	To	Reason	Status	Applied
1	2025-11-08T18:30:00.000Z	2025-11-16T18:30:00.000Z	need to go home	pending	09/11/2025, 22:22:10

Notices

Tomorrow is Holiday

Tomorrow is holiday

Audience: All hostels

09/11/2025, 22:25:56

Conclusion

The system successfully manages hostel-related student data using a structured relational database.

Admin can add students and post notices efficiently.

Students can log in, view their profiles, and submit maintenance/leave requests where their hostel is **securely auto-assigned**, preventing tampering.

The design is maintainable, scalable, and can be enhanced further with payment modules, warden logins, or biometric entry logs.

References

- <https://dev.mysql.com/doc/>
- <https://expressjs.com/>
- <https://nodejs.org/>
- W3Schools: HTML/CSS/JS Tutorials
- Classroom DBMS lecture notes